Nullspace

Null(A) → contain all the solution (set of vectors) for which $Ax=0$

Condition
1) closed under multiplication
2) closed under Addition

$$A=\begin{bmatrix}1&2&2&2\\2&4&6&8\\3&6&8&10\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix}=0$$

x

$$\begin{bmatrix}\boxed{1}&2&2&2\\0&\boxed{0}&\boxed{2}&4\\0&0&2&4\end{bmatrix}\rightarrow\begin{bmatrix}\boxed{1}&2&\boxed{2}&\boxed{2}\\0&0&\boxed{2}&4\\0&0&0&\boxed{0}\end{bmatrix}$$

pivot cols = 2 = rank

→ gauss stop

Free cols. (can be anything)

rank(A) = 2

↳ # of pivots

$Ax=0$

$Ux=0$

☆ heart of today's class:

$$1x_1+2x_2+2x_3+\textcircled{2}x_4=0$$

solve both

$$2x_3+\textcircled{4}x_4=0$$

$0=0$ → nothing new cross it.

$x_2=0$ $\quad x_2=1$

$x_4=1$ $\quad x_4=0$

$$Ax=\begin{bmatrix}2\\0\\-2\\1\end{bmatrix}=0$$

1st col, 2nd col

could be any value

X 3rd col X 4th col

choosing 0 & 1, → $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ Kinda Identity

machines can never do back substitution.
so mimic it.

got only 2

$$\left\{ c\begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix} + d\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

surely within Nullspace
still there could be caveats

basis of Nullspace → take any combination
|| & Multiply → gives 0.
All ||

special solution
→ 0, 1 ?
take linear combination

if $x_1, x_2, x_3 \rightarrow$ $\begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}$

$$\begin{bmatrix} \boxed{1} & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \\ 2 & 8 & 10 \end{bmatrix} \rightarrow \begin{bmatrix} \boxed{1} & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 4 & 4 \end{bmatrix}$$

2 rank

$$\rightarrow \begin{bmatrix} \boxed{1} & 2 & 3 \\ 0 & \boxed{2} & 2 \\ 0 & 4 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$



At A rank = 2
AT U = 2

$x_1 + 2x_2 + 3x_3 = 0$ → free var

$2x_2 + 2x_3 = 0$

$0 = 0$
$0 = 0$

$2x_2 = -2(1)$
$x_2 = -1$

$x_1 = 2 - 3$
$= -1$

$x_3 = 1$

(taking 0 same problm arises

$c \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$

$c = -1$
1, 1, -1 sol

↳ A line in $\mathbb{R}^3$ → 1 linearly independent